Bandersnatch VRF-AD Specification

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Abstract

This specification defines three Verifiable Random Function with Additional Data (VRF-AD) schemes – IETF VRF, Thin VRF, and Pedersen VRF – built on a transcript-based Fiat-Shamir transform with support for multiple input/output pairs via delinearization. The IETF VRF extends RFC-9381 [1]; the Thin VRF and Pedersen VRF follow the constructions introduced by BCHSV23 [2], with the Pedersen VRF serving as a building block for anonymized ring signatures as described in VG24 [3]. All schemes are instantiated over the Bandersnatch elliptic curve, constructed over the BLS12-381 scalar field as specified in MSZ21 [4].

1. Preliminaries

1.1. Groups and Fields

- $\mathbf{G}$: Bandersnatch curve cyclic group of prime order r.
- $\mathbf{F}$: Scalar field of prime order r (i.e. $\mathbb{Z}_r$).
- $\mathbf{\Sigma}^k$: Octet strings with length $k \in \mathbb{N}$ ($*$ for arbitrary length).
- $\mathcal{O}$: Identity point of $\mathbf{G}$.

The EC group $\mathbf{G}$ is the prime subgroup of the Bandersnatch elliptic curve, in Twisted Edwards form, with finite field and curve parameters as specified in MSZ21 [4]. For this group, `fLen` = `qLen` = 32 and `cofactor` = 4.

- $G \in \mathbf{G}$: Prime order group generator.

  G_x=18886178867200960497001835917649091219057080094937609519140440539760939937304

  G_y=19188667384257783945677642223292697773471335439753913231509108946878080696678

  - Compressed: `0x664197ccb667315e6064e4ee81ad8c3586d5dcba508b7d150f3e12da9e666c2a`

1.2. Notation

- $x \in \mathbf{F}$: Secret key scalar.
- $Y \in \mathbf{G}$: Public key point defined as $x \cdot G$.
- $i \in \mathbf{\Sigma}^*$: VRF input data.
- $I \in \mathbf{G}$: VRF input point.
- $O \in \mathbf{G}$: VRF output point.
- $o \in \mathbf{\Sigma}^k$: VRF output hash.
- T: Transcript state (section 1.10).

1.3. Encoding

- `serialize`(P): Encodes a point in compressed form. The y coordinate is serialized in little-endian and the most significant bit of the last octet encodes the sign of x. This gives `ptLen` = `fLen` = 32.
- `serialize`(s): Encodes a scalar into 32 octets in little-endian representation.
- `le32`(n): Encode integer n as a 4-byte little-endian octet string.
- `from_le_bytes_mod_order`(buf): Interpret octet string buf as a little-endian integer and reduce modulo the prime field order r.

Point deserialization MUST output "INVALID" if the octet-string does not decode to a point on the prime subgroup $\mathbf{G}$.

1.4. Constants

- `suite_string` = `"Bandersnatch_SHA-512_ELL2"`.
- `encode_to_curve_salt` = `""` (empty - no salt).
- `challenge_len` = 16 bytes (128-bit security).
- `expanded_scalar_len` = $\lceil(\lceil\log_2(r)\rceil + 128)/8\rceil$ = 48 bytes for Bandersnatch.

Domain separation tags used throughout the protocol:

Tag	Value	Usage
Challenge	0x01	Challenge derivation
NonceExpand	0x02	Nonce secret expansion
Nonce	0x03	Nonce derivation
PointToHash	0x04	VRF output hashing
Delinearize	0x05	Delinearization scalars
PedersenBlinding	0x80	Pedersen blinding factor

Tag	Value	Usage

1.5. Secret Key Generation

Implementations should provide a method for deterministic secret generation from seeds. One RECOMMENDED method is described in Appendix A. However, any secure method that outputs uniformly random scalars in $\mathbf{F}$ is acceptable.

1.6. VRF Input

The VRF input point $I \in \mathbf{G}$ is derived from the input octet-string using the `hash_to_curve` method defined in section 3 of RFC-9380 [5], instantiated with the *Elligator 2* map to curve (section 6.8.2) and `expand_message_xmd` with SHA-512 (section 5.3.1).

This is the random oracle (`_RO_`) construction: the input is hashed to two independent field elements, each is mapped to a curve point via Elligator 2, and the results are added.

The domain separation tag is:

$$DST = \text{"ECVRF_"} \parallel \texttt{h2c_suite_id} \parallel \texttt{suite_string}$$

where `h2c_suite_id` = `"Bandersnatch_XMD:SHA-512_ELL2_RO_"` and `suite_string` = `"Bandersnatch_SHA-512_ELL2"` (section 1.4).

$$I \leftarrow \texttt{hash_to_curve}(i)$$

1.7. VRF Output

The VRF output point is generated from the VRF input point and secret key scalar:

$$O \leftarrow x \cdot I$$

The VRF output hash is a fixed-length octet string derived from the output point using a transcript-based point-to-hash procedure.

Input:

- $O \in \mathbf{G}$: VRF output point.
- $N \in \mathbb{N}$: Desired output length in bytes.

Output:

- $o \in \mathbf{\Sigma}^N$: VRF output hash.

Steps:

1. $T \leftarrow \texttt{new}(\text{SUITE_ID})$
2. $T.\texttt{absorb}(\texttt{PointToHash})$
3. $T.\texttt{absorb}(\texttt{serialize}(O))$
4. $o \leftarrow T.\texttt{squeeze}(N)$

1.8. Additional Data

An arbitrary length octet-string provided by the user to be signed together with the generated VRF output. This data doesn't influence the produced VRF output.

1.9. VRF-AD

Regardless of the specific scheme, a *Verifiable Random Function with Additional Data (VRF-AD)* can be concisely represented by three primary functions:

- $prove(x, \overline{io}, ad) \mapsto \Pi$
- $verify(Y, \overline{io}, ad, \Pi) \mapsto (\top \mid \bot)$
- $output(O) \mapsto o$

Where $\overline{io}$ is a sequence of (I_i, O_i) input/output pairs.

1.10. Transcript

The transcript provides a Fiat-Shamir transform with an absorb/squeeze interface. Data is absorbed into an internal hash state; output bytes are squeezed from it. After the first squeeze, no further absorbs are permitted.

Abstract interface:

- $\texttt{new}(label)$: Initialize with a domain-separation label.
- $\texttt{absorb}(data)$: Feed bytes into the hash state. Must not be called after squeeze.
- $\texttt{squeeze}(n) \rightarrow \mathbf{\Sigma}^n$: Produce n output bytes.
- $\texttt{fork}()$: Clone the transcript state.

Concrete construction (`HashTranscript<SHA-512>`):

Initialization: $\texttt{new}(label)$ creates a fresh SHA-512 state and feeds $\texttt{le32}(\texttt{len}(label)) \parallel label$ into it.

Absorb: feeds raw bytes directly into the SHA-512 state.

Squeeze (counter-mode XOF): on the first squeeze call, finalize the SHA-512 state to obtain a 64-byte *seed*. Then produce output blocks:

$$block_i = \text{SHA-512}(seed \parallel \texttt{le32}(i)) \quad \text{for } i = 0, 1, 2, \ldots$$

Each block yields 64 bytes. Output is read sequentially across blocks; partial block state is preserved between squeeze calls.

Fork: duplicates the full internal state (including any partial block position if squeezing has begun).

1.11. VRF Transcript

Shared transcript construction used by all VRF-AD schemes. Absorbs input/output pairs, derives delinearization scalars, merges pairs into a single pair, and absorbs additional data.

Input:

- $\overline{io} \in (\mathbf{G} \times \mathbf{G})^n$: Sequence of input/output pairs.
- $ad \in \mathbf{\Sigma}^*$: Additional data octet-string.

Output:

- T: Transcript state (with ad absorbed).
- $(I_m, O_m) \in \mathbf{G} \times \mathbf{G}$: Merged input/output pair.

Steps:

1. $T \leftarrow \texttt{new}(\text{SUITE_ID})$
2. For each (I_i, O_i) in $\overline{io}$: $T.\texttt{absorb}(\texttt{serialize}(I_i) \parallel \texttt{serialize}(O_i))$
3. Delinearize:
   - If $n = 0$: $(I_m, O_m) \leftarrow (\mathcal{O}, \mathcal{O})$
   - If $n = 1$: $(I_m, O_m) \leftarrow (I_0, O_0)$
   - If $n \geq 2$: $T' \leftarrow T.\texttt{fork}()$, $T'.\texttt{absorb}(\texttt{Delinearize} \parallel \texttt{le32}(n))$, squeeze n scalars $z_i \leftarrow \texttt{from_le_bytes_mod_order}(T'.\texttt{squeeze}(\texttt{challenge_len}))$, $I_m \leftarrow \sum_i z_i \cdot I_i$, $O_m \leftarrow \sum_i z_i \cdot O_i$
4. $T.\texttt{absorb}(\texttt{le32}(\texttt{len}(ad)) \parallel ad)$
5. Return $(T, (I_m, O_m))$

1.12. Nonce Procedure

Deterministic nonce generation inspired by [RFC-8032] section 5.1.6. The transcript carries shared state from `vrf_transcript`, binding the nonce to the I/O pairs and additional data.

Input:

- $sk \in \mathbf{F}$: Secret scalar.
- T: Transcript state (consumed).

Output:

- $k \in \mathbf{F}$: Nonce scalar.

Steps:

1. $T' \leftarrow T.\texttt{fork}()$
2. $T'.\texttt{absorb}(\texttt{NonceExpand} \parallel \texttt{serialize}(sk))$
3. $h \leftarrow T'.\texttt{squeeze}(64)$
4. $T.\texttt{absorb}(\texttt{Nonce} \parallel h[32..64])$
5. $k \leftarrow \texttt{from_le_bytes_mod_order}(T.\texttt{squeeze}(\text{expanded_scalar_len}))$

Note: T is consumed (mutated then squeezed). Callers must pass forks where the transcript is needed afterwards.

1.13. Challenge Procedure

Derives a challenge scalar by absorbing curve points into the transcript and squeezing.

Input:

- $\bar{P} \in \mathbf{G}^m$: Sequence of m points.
- T: Transcript state (consumed).

Output:

- $c \in \mathbf{F}$: Challenge scalar.

Steps:

1. $T.\texttt{absorb}(\texttt{Challenge})$
2. For each P_i in $\bar{P}$: $T.\texttt{absorb}(\texttt{serialize}(P_i))$
3. $c \leftarrow \texttt{from_le_bytes_mod_order}(T.\texttt{squeeze}(\texttt{challenge_len}))$

2. IETF VRF

Based on IETF RFC-9381 which is extended with a transcript-based Fiat-Shamir transform, support for additional data (ad), and multiple I/O pairs via delinearization.

2.1. Prove

Input:

- $x \in \mathbf{F}$: Secret key.
- $\overline{io} \in (\mathbf{G} \times \mathbf{G})^n$: VRF input/output pairs.
- $ad \in \mathbf{\Sigma}^*$: Additional data octet-string.

Output:

- $\pi = (c, s) \in (\mathbf{F}, \mathbf{F})$: Schnorr-like proof.

Steps:

1. $Y \leftarrow x \cdot G$
2. $(T, (I_m, O_m)) \leftarrow \texttt{vrf_transcript}(\overline{io}, ad)$
3. $k \leftarrow \texttt{nonce}(x, T.\texttt{fork}())$
4. $k_b \leftarrow k \cdot G, \quad k_h \leftarrow k \cdot I_m$
5. $c \leftarrow \texttt{challenge}([Y, k_b, k_h], T)$
6. $s \leftarrow k + c \cdot x$
7. $\pi \leftarrow (c, s)$

The challenge scalar c is serialized as the first `challenge_len` bytes of the little-endian representation.

2.2. Verify

Input:

- $Y \in \mathbf{G}$: Public key.
- $\overline{io} \in (\mathbf{G} \times \mathbf{G})^n$: VRF input/output pairs.
- $ad \in \mathbf{\Sigma}^*$: Additional data octet-string.
- $\pi = (c, s) \in (\mathbf{F}, \mathbf{F})$: Schnorr-like proof.

Output:

- $\theta \in \{\top, \bot\}$: $\top$ if proof is valid, $\bot$ otherwise.

Steps:

1. Validate all points $\in \mathbf{G}$, output $\bot$ if invalid.
2. $(T, (I_m, O_m)) \leftarrow \texttt{vrf_transcript}(\overline{io}, ad)$
3. $U \leftarrow s \cdot G - c \cdot Y$
4. $V \leftarrow s \cdot I_m - c \cdot O_m$

5. $c' \leftarrow \texttt{challenge}([Y, U, V], T)$
6. $\theta \leftarrow \top$ if $c = c'$ else $\bot$

3. Thin VRF

Thin VRF is derived from the PedVRF construction in section 4 of BCHSV23 [2], reduced to EC-VRF form by setting the blinding factor $b = 0$ and using $pk = sk \cdot G$ directly (see remark on page 13 of the paper).

The scheme merges the Schnorr key pair (G, Y) and VRF I/O pairs into a single DLEQ relation via delinearization, then proves it with a Schnorr-like proof (R, s). Storing the nonce commitment R (rather than the challenge c) enables batch verification.

Security: VRF input points MUST be constructed via hash-to-curve. If a prover knows d such that $I = d \cdot G$, they can forge arbitrary outputs for that input, because the delinearization merges the Schnorr and VRF pairs into a single check that collapses when all points are multiples of G.

3.1. Prove

Input:

- $x \in \mathbf{F}$: Secret key.
- $\overline{io} \in (\mathbf{G} \times \mathbf{G})^n$: VRF input/output pairs.
- $ad \in \mathbf{\Sigma}^*$: Additional data octet-string.

Output:

- $\pi = (R, s) \in (\mathbf{G}, \mathbf{F})$: Thin VRF proof.

Steps:

1. $Y \leftarrow x \cdot G$
2. $\overline{io}' \leftarrow [(G, Y)] \parallel \overline{io}$ (prepend Schnorr pair)
3. $(T, (I_m, O_m)) \leftarrow \texttt{vrf_transcript}(\overline{io}', ad)$
4. $k \leftarrow \texttt{nonce}(x, T.\texttt{fork}())$
5. $R \leftarrow k \cdot I_m$
6. $c \leftarrow \texttt{challenge}([R], T)$
7. $s \leftarrow k + c \cdot x$
8. $\pi \leftarrow (R, s)$

3.2. Verify

Input:

- $Y \in \mathbf{G}$: Public key.
- $\overline{io} \in (\mathbf{G} \times \mathbf{G})^n$: VRF input/output pairs.
- $ad \in \mathbf{\Sigma}^*$: Additional data octet-string.
- $\pi = (R, s) \in (\mathbf{G}, \mathbf{F})$: Thin VRF proof.

Output:

- $\theta \in \{\top, \bot\}$: $\top$ if proof is valid, $\bot$ otherwise.

Steps:

1. Validate all points $\in \mathbf{G}$, output $\bot$ if invalid.
2. $\overline{io}' \leftarrow [(G, Y)] \parallel \overline{io}$
3. $(T, (I_m, O_m)) \leftarrow \texttt{vrf_transcript}(\overline{io}', ad)$
4. $c \leftarrow \texttt{challenge}([R], T)$
5. $\theta \leftarrow \top$ if $s \cdot I_m = R + c \cdot O_m$ else $\bot$

4. Pedersen VRF

Pedersen VRF resembles IETF EC-VRF but replaces the public key with a Pedersen commitment to the secret key, which makes this VRF useful in anonymized ring proofs.

The scheme proves that the output has been generated with a secret key associated with a blinded public key (instead of the public key). The blinded public key is a cryptographic commitment to the public key, and it can be unblinded to prove that the output of the VRF corresponds to the public key of the signer.

This specification mostly follows the design proposed by BCHSV23 [2] in section 4 with some details about blinding base point value and challenge generation procedure.

The *blinding base* $B \in \mathbf{G}$ is defined as follows:

$$B_x = 15502292510512467136772963637174542389568776133586142241717400964712787983 12$$

$$B_y = 28442734166467795856797249030329035618871580593056783094884474814923353898473$$

- Compressed: `0xe93da06b869766b158d20b843ec648cc68e0b7ba2f7083acf0f154205d04e23e`

4.1. Prove

Input:

- $x \in \mathbf{F}$: Secret key.
- $\overline{io} \in (\mathbf{G} \times \mathbf{G})^n$: VRF input/output pairs.
- $ad \in \mathbf{\Sigma}^*$: Additional data octet-string.

Output:

- $\pi = (\bar{Y}, R, O_k, s, s_b) \in (\mathbf{G}, \mathbf{G}, \mathbf{G}, \mathbf{F}, \mathbf{F})$: Pedersen proof.
- $b \in \mathbf{F}$: Blinding factor.

Steps:

1. $(T, (I_m, O_m)) \leftarrow \texttt{vrf_transcript}(\overline{io}, ad)$
2. $b \leftarrow \texttt{blinding}(x, T.\texttt{fork}())$ (see Appendix A.2)
3. $\bar{Y} \leftarrow x \cdot G + b \cdot B$
4. $T_k \leftarrow T.\texttt{fork}(), \quad T_k.\texttt{absorb}(\texttt{serialize}(b)), \quad k \leftarrow \texttt{nonce}(x, T_k)$
5. $T_{kb} \leftarrow T.\texttt{fork}(), \quad T_{kb}.\texttt{absorb}(\texttt{serialize}(x)), \quad k_b \leftarrow \texttt{nonce}(b, T_{kb})$
6. $R \leftarrow k \cdot G + k_b \cdot B$
7. $O_k \leftarrow k \cdot I_m$
8. $c \leftarrow \texttt{challenge}([\bar{Y}, R, O_k], T)$
9. $s \leftarrow k + c \cdot x$
10. $s_b \leftarrow k_b + c \cdot b$
11. $\pi \leftarrow (\bar{Y}, R, O_k, s, s_b)$

The nonce cross-binding is critical: k is bound to b (step 4) and k_b is bound to x (step 5). This prevents secret/blinding recovery from two proofs with the same (secret, input, ad) but different blinding factors.

4.2. Verify

Input:

- $\overline{io} \in (\mathbf{G} \times \mathbf{G})^n$: VRF input/output pairs.
- $ad \in \mathbf{\Sigma}^*$: Additional data octet-string.
- $\pi = (\bar{Y}, R, O_k, s, s_b) \in (\mathbf{G}, \mathbf{G}, \mathbf{G}, \mathbf{F}, \mathbf{F})$: Pedersen proof.

Output:

- $\theta \in \{\top, \bot\}$: $\top$ if proof is valid, $\bot$ otherwise.

Steps:

1. Validate all points $\in \mathbf{G}$, output $\perp$ if invalid.
2. $(T, (I_m, O_m)) \leftarrow$ `vrf_transcript`$(\overline{io}, ad)$
3. $c \leftarrow$ `challenge`$([\bar{Y}, R, O_k], T)$
4. $\theta_0 \leftarrow \top$ if $O_k + c \cdot O_m = s \cdot I_m$ else $\perp$
5. $\theta_1 \leftarrow \top$ if $R + c \cdot \bar{Y} = s \cdot G + s_b \cdot B$ else $\perp$
6. $\theta = \theta_0 \wedge \theta_1$

Note: no public key appears in the verify inputs – verification uses the committed key $\bar{Y}$ from the proof.

5. Ring VRF

Anonymized ring VRF based on Pedersen VRF (section 4) and Ring Proof as proposed in VG24.

The following configuration specializes VG24 for the concrete scheme:

- **Groups and Fields**:
  - $\mathbb{G}_{\mathbb{K}}$: BLS12-381 prime order subgroup.
  - $\mathbb{F}$: BLS12-381 scalar field.
  - J: Bandersnatch curve defined over $\mathbb{F}$.
- **Polynomial Commitment Scheme**
  - KZG with SRS derived from Zcash powers of tau ceremony.
- **Fiat-Shamir Transform**
  - `ark-transcript`.
  - Begin with empty transcript and "ring-proof" label.
  - Push R to the transcript after instancing.
  - TODO: Specify the order and how parameters are added to the transcript as we progress the protocol.
- Accumulator seed point in Twisted Edwards form:

  S_x=37805570861274048643170021838972902516980894313648523898085159469000338764576

  S_y=14738305321141000190236674389841754997202271418876976886494444739226156422510
  - Compressed: `0x6e5574f9077fb76c885c36196a832dbadd64142d305be5487724967acf9595a0`
- Padding point in Twisted Edwards form:

  $\square_x$=26287722405578650394504321825321286533153045350760430979437739593351290020913

  $\square_y$=19058981610000167534379068105702216971787064146691007947119244515951752366738

  - Compressed: `0x92ca79e61dd90c1573a8693f199bf6e1e86835cc715cdcf93f5ef222560023aa`

  A point with unknown discrete logarithm derived using the `ECVRF_encode_to_curve` function as described in section 1.6 with input the string: `"ring-proof-pad"`.

- Polynomials domain ($\langle\omega\rangle = \mathbb{D}$) generator:

  $$\omega = 49307615728544765012166121802278658070711169839041683575071795236746050763237$$

- $|\mathbb{D}| = 2048$

5.1. Prove

Input:

- $x \in \mathbf{F}$: Secret key.
- $\overline{io} \in (\mathbf{G} \times \mathbf{G})^n$: VRF input/output pairs.
- $ad \in \mathbf{\Sigma}^*$: Additional data octet-string.
- P: Ring prover (encapsulates ring keys and prover index).

Output:

- $\pi_p \in (\mathbf{G}, \mathbf{G}, \mathbf{G}, \mathbf{F}, \mathbf{F})$: Pedersen proof.
- $\pi_r \in ((G_1)^4, (\mathbf{F})^7, G_1, \mathbf{F}, G_1, G_1)$: Ring proof.

Steps:

1. $(\pi_p, b) \leftarrow Pedersen.prove(x, \overline{io}, ad)$
2. $\pi_r \leftarrow Ring.prove(P, b)$

The blinding factor b is derived internally by Pedersen prove (section 4.2, step 2) and forwarded to the ring prover.

5.2. Verify

Input:

- $\overline{io} \in (\mathbf{G} \times \mathbf{G})^n$: VRF input/output pairs.
- $ad \in \mathbf{\Sigma}^*$: Additional data octet-string.
- $\pi_p \in (\mathbf{G}, \mathbf{G}, \mathbf{G}, \mathbf{F}, \mathbf{F})$: Pedersen proof.
- $\pi_r \in ((G_1)^4, (\mathbf{F})^7, G_1, \mathbf{F}, G_1, G_1)$: Ring proof.
- $V \in (G_1)^3$: Ring verifier (pre-processed commitment).

Output:

- $\theta \in \{\top, \bot\}$: $\top$ if proof is valid, $\bot$ otherwise.

Steps:

1. $\theta_0 \leftarrow Pedersen.verify(\overline{io}, ad, \pi_p)$
2. $(\bar{Y}, R, O_k, s, s_b) \leftarrow \pi_p$
3. $\theta_1 \leftarrow Ring.verify(V, \pi_r, \bar{Y})$
4. $\theta \leftarrow \theta_0 \wedge \theta_1$

Appendix A. Recommendations

A.1. Deterministic Secret Key Scalar Generation

For convenience and to facilitate deterministic key generation (e.g., from mnemonic phrases), we suggest the following method to derive a secret scalar from an arbitrary byte string seed. This procedure is not mandated by the specification and may be replaced by any secure method that produces uniformly distributed scalars in the field $\mathbf{F}$.

Input:

- $seed \in \mathbf{\Sigma}^*$: seed octet-string.

Output:

- $secret \in \mathbf{F}$: secret key scalar.

Steps:

1. $cnt \leftarrow 0$
2. $T \leftarrow \texttt{new}$("ark-vrf-keygen")
3. $T.\texttt{absorb}(seed)$
4. If $cnt > 0$: $T.\texttt{absorb}([cnt])$
5. $secret \leftarrow \texttt{nonce}(0, T)$
6. If $secret = 0$: increment cnt and go to step 2
7. Return $secret$

The `nonce` procedure (section 1.12) is called with a zero scalar and provides domain separation, uniform scalar derivation, and negligible modular reduction bias.

A.2. Deterministic Blinding Factor Generation

For Pedersen VRF, the blinding factor may be generated deterministically from the secret key and the VRF transcript state. This procedure uses the nonce function (section 1.12) with a distinct domain separator.

Input:

- $sk \in \mathbf{F}$: Secret scalar.
- T: Transcript state (from `vrf_transcript`).

Output:

- $b \in \mathbf{F}$: Blinding factor scalar.

Steps:

1. T.`absorb`(`PedersenBlinding`)
2. $b \leftarrow$ `nonce`(sk, T)

Appendix B. Test Vectors

TODO: The test vectors below are stale and must be regenerated after the nonce procedure change (section 1.10) is applied to the reference implementation.

The test vectors in this section were generated using `ark-vrf` libraries revision `bf2d1cf`.

B.1. IETF VRF Test Vectors

Schema:

```
sk (x): Secret key,
pk (Y): Public key,
in (alpha): Input octet-string,
salt: encode_to_curve_salt (empty for this suite),
ad: Additional data octet-string,
h (I): VRF input point,
gamma (O): VRF output point,
out (beta): VRF output octet string,
proof_c: Proof 'c' component,
proof_s: Proof 's' component,
```

Vector 1

```
3d6406500d4009fdf2604546093665911e753f2213570a29521fd88bc30ede18,
a1b1da71cc4682e159b7da23050d8b6261eb11a3247c89b07ef56ccd002fd38b,
-,
-,
c5eaf38334836d4b10e05d2c1021959a917e08eaf4eb46a8c4c8d1bec04e2c00,
e7aa5154103450f0a0525a36a441f827296ee489ef30ed8787cff8df1bef223f,
fdeb377a4ffd7f95ebe48e5b43a88d069ce62188e49493500315ad55ee04d744
..2b93c4c91d5475370e9380496f4bc0b838c2483bce4e133c6f18b0adbb9e4722,
439fd9495643314fa623f2581f4b3d7d6037394468084f4ad7d8031479d9d101,
```

```
828bedd2ad95380b11f67a05ea0a76f0c3fef2bee9f043f4dffdddde09f55c01,
```

Vector 2

```
8b9063872331dda4c3c282f7d813fb3c13e7339b7dc9635fdc764e32cc57cb15,
5ebfe047f421e1a3e1d9bbb163839812657bbb3e4ffe9856a725b2b405844cf3,
0a,
-,
8c1d1425374f01d86b23bfeab770c60b58d2eeb9afc5900c8b8a918d09a6086b,
60f32f5ad3e9694b82ccc0a735edb2f940f757ab333cc5f7b0a41158b80f574f,
44f3728bc5ad550aeeb89f8db340b2fceffc946be3e2d8c5d99b47c1fce344b3
..c7fcee223a9b29a64fe4a86a9994784bc165bb0fba03ca0a493f75bee89a0946,
8aa1c755a00a6a25bdecda197ee1b60a01e50787bd10aa976133f4c39179330e,
18c74ffd67e6abc658e2d05ecd3101ddc0c33623823f2395538cf8d39e654f12,
```

Vector 3

```
6db187202f69e627e432296ae1d0f166ae6ac3c1222585b6ceae80ea07670b14,
9d97151298a5339866ddd3539d16696e19e6b68ac731562c807fe63a1ca49506,
-,
0b8c,
c5eaf38334836d4b10e05d2c1021959a917e08eaf4eb46a8c4c8d1bec04e2c00,
67a348e256d908eb695d15ee0d869efef2bcf9f0fea646e788f967abbc0464dd,
edde0178045133eb03ef4d1ad8b978a56ee80ec4eab8830d6bc6c08003138841
..6657d3c449d9398cc4385d1c8a2bb19bcf61ff086e5a6c477a0302ce270d1abf,
aec4d1cf308cb4cb400190350e69f4fb309255aa738fff5a6ac4ced7538fce03,
54e5d38a76f309ce63ca82465160abd8d75b78805a0b499e60c26436de4a8e01,
```

Vector 4

```
b56cc204f1b6c2323709012cb16c72f3021035ce935fbe69b600a88d842c7407,
dc2de7312c2850a9f6c103289c64fbd76e2ebd2fa8b5734708eb2c76c0fb2d99,
73616d706c65,
-,
672e8c7a8e6d3eca67df38f11d50f3d7dbb26fa8e27565a5424e6f8ac4555dcc,
4d3e0524fc59374f1fdad8e471c695469b45ecf69c1de85c6c1230e888dd4cbe,
36127f8aee7c61048984f0a208bf6d334db9dacbeeeef9ff2d17117e81232832
..1462eb3ef602f5911d77ab11f815eb4154ba95c934e414198ef000a61b4de31a,
b72598f235145a377911caa794ba85820173c4c49b7be3b05d847b2c753e0311,
e8e34ad3131388a88eb7f80bd874f3421c378d4ad45911c4bc16e4cdc17b5716,
```

Vector 5

```
da36359bf1bfd1694d3ed359e7340bd02a6a5e54827d94db1384df29f5bdd302,
decb0151cbeb49f76f10419ab6a96242bdc87baac8a474e5161123de4304ac29,
42616e646572736e6174636820766563746f72,
-,
```

```
4315192d2ce9e52ceb449a6b4da7f7e6636e53592c7f5e236763e21e9bac24c7,
9508104b820469687488d83f729288d9f70fc0523318beff44a47da10d490b3c,
4ee61f3c000544aa48c565e143e05c6501a623bdbf02a0a408b97433660b4907
..715f75890cc0e45cdd7116e3da15b15c3c637782e8e05d05c0d5895e5fe583d1,
ad6af59b4b84f18187c694ef374687d13517cb53508ff9dafa37d0c759e9601c,
4c1269d9d161dabd082fc606af979eca7f6c3ab68e78261dc6fb9fbbb98c9704,
```

Vector 6

```
da36359bf1bfd1694d3ed359e7340bd02a6a5e54827d94db1384df29f5bdd302,
decb0151cbeb49f76f10419ab6a96242bdc87baac8a474e5161123de4304ac29,
42616e646572736e6174636820766563746f72,
1f42,
4315192d2ce9e52ceb449a6b4da7f7e6636e53592c7f5e236763e21e9bac24c7,
9508104b820469687488d83f729288d9f70fc0523318beff44a47da10d490b3c,
4ee61f3c000544aa48c565e143e05c6501a623bdbf02a0a408b97433660b4907
..715f75890cc0e45cdd7116e3da15b15c3c637782e8e05d05c0d5895e5fe583d1,
4fa53519bd9d17acae4d1021416557d11b84dd4670b563770c14eb98161eaa08,
0f7f9bee9077427f547e69b919cf8d63823c14b20085fd9516768e0f5e3d3f0e,
```

Vector 7

```
35b877a25c394512292b82bdf8468e98eaf03c79c7fc9d53546dadc5fb75b500,
b0e1f208f9d6e5b310b92014ea7ef3011e649dab038804759f3766e01029d623,
42616e646572736e6174636820766563746f72,
1f42,
4315192d2ce9e52ceb449a6b4da7f7e6636e53592c7f5e236763e21e9bac24c7,
6d1dd583bea262323c7dc9e94e57a472e09874e435719010eeafae503c433f16,
09106f062ac07846f3f841f64765527b333575143483855d633f99ccc2e8e306
..e6239ff79a1272cff931e8d0ac6c390328486329118ad40a18b85184da1837ff,
6dbeeab9648505fa6a95de52d611acfbb2febacc58cdc7d0ca45abd8c952ef12,
ce7f4a2354a6c3f97aee6cc60c6aa4c4430b12ed0f0ef304b326c776618d7609,
```

B.2. Pedersen VRF Test Vectors

Schema:

```
sk (x): Secret key,
pk (Y): Public key,
in (alpha): Input octet-string,
salt: encode_to_curve_salt (empty for this suite),
ad: Additional data octet-string,
h (I): VRF input point,
gamma (O): VRF output point,
out (beta): VRF output octet string,
blinding: Blinding factor,
proof_pk_com (Y^-): Public key commitment,
```

```
proof_r: Proof 'R' component,
proof_ok: Proof 'O_k' component,
proof_s: Proof 's' component,
proof_sb: Proof 's_b' component
```

Vector 1


```
3d6406500d4009fdf2604546093665911e753f2213570a29521fd88bc30ede18,
a1b1da71cc4682e159b7da23050d8b6261eb11a3247c89b07ef56ccd002fd38b,
-,
-,
-,
c5eaf38334836d4b10e05d2c1021959a917e08eaf4eb46a8c4c8d1bec04e2c00,
e7aa5154103450f0a0525a36a441f827296ee489ef30ed8787cff8df1bef223f,
fdeb377a4ffd7f95ebe48e5b43a88d069ce62188e49493500315ad55ee04d744
..2b93c4c91d5475370e9380496f4bc0b838c2483bce4e133c6f18b0adbb9e4722,
01371ac62e04d1faaadbebaa686aaf122143e2cda23aacbaa4796d206779a501,
3b21abd58807bb6d93797001adaacd7113ec320dcf32d1226494e18a57931fc4,
8123054bfdb6918e0aa25c3337e6509eea262282fd26853bf7cd6db234583f5e,
ac57ce6a53a887fc59b6aa73d8ff0e718b49bd9407a627ae0e9b9e7c5d0d175b,
0d379b65fb1e6b2adcbf80618c08e31fd526f06c2defa159158f5de146104c0f,
e2ca83136143e0cac3f7ee863edd3879ed753b995b1ff8d58305d3b1f323630b,
```


Vector 2


```
8b9063872331dda4c3c282f7d813fb3c13e7339b7dc9635fdc764e32cc57cb15,
5ebfe047f421e1a3e1d9bbb163839812657bbb3e4ffe9856a725b2b405844cf3,
0a,
-,
-,
8c1d1425374f01d86b23bfeab770c60b58d2eeb9afc5900c8b8a918d09a6086b,
60f32f5ad3e9694b82ccc0a735edb2f940f757ab333cc5f7b0a41158b80f574f,
44f3728bc5ad550aeeb89f8db340b2fceffc946be3e2d8c5d99b47c1fce344b3
..c7fcee223a9b29a64fe4a86a9994784bc165bb0fba03ca0a493f75bee89a0946,
99ff52abf49d67c4303ac4a8a00984d04c06388f5f836ebd37031f0e76245815,
c1322e7a65b83996c25e37a84e36598333b0d417619242c0cb3d9d972edde848,
7a4363e0bf9cd18317287d681ab05704982b0088ce373f696dbdf3909a902b36,
fc8770c209212640742d53e2f40e5c30fffae574f90fdc670ff11a1127586c03,
93f7c9d73eec05e500b758f645a2967e62b2206e57eff5f9b99bfc71812e620d,
c864de36e0b428f6fb4ef470f94ec9601716cb26ad96f3359e4a1ec110794a0b,
```


Vector 3


```
6db187202f69e627e432296ae1d0f166ae6ac3c1222585b6ceae80ea07670b14,
9d97151298a5339866ddd3539d16696e19e6b68ac731562c807fe63a1ca49506,
-,
-,
```

```
0b8c,
c5eaf38334836d4b10e05d2c1021959a917e08eaf4eb46a8c4c8d1bec04e2c00,
67a348e256d908eb695d15ee0d869efef2bcf9f0fea646e788f967abbc0464dd,
edde0178045133eb03ef4d1ad8b978a56ee80ec4eab8830d6bc6c08003138841
..6657d3c449d9398cc4385d1c8a2bb19bcf61ff086e5a6c477a0302ce270d1abf,
e22ec3e4a2a4132237eb8a62bcc5ed864593cfde08e53b1632ecd3245761c808,
54c04f259f9e40ee086031d29960b12b6b6407e9de14985001c7265587941831,
9200b650a0c20b0ef73ccd7651ffc7af154e5e02879dc8666025c245aa547f01,
35f8dc0f744d1850513c46b6b4640716cbb4643da26cfe67f8c701486e0b4cae,
5faa89369589174f4202d6e53e8b4ef10a49b2ad8face60d7cb28bfc8f43bf0e,
017093ff8d22ba2f3852141365a1452fbb5ab8cf6f20cb04555e3163f8d88f13,
```

Vector 4

```
b56cc204f1b6c2323709012cb16c72f3021035ce935fbe69b600a88d842c7407,
dc2de7312c2850a9f6c103289c64fbd76e2ebd2fa8b5734708eb2c76c0fb2d99,
73616d706c65,
-,
-,
672e8c7a8e6d3eca67df38f11d50f3d7dbb26fa8e27565a5424e6f8ac4555dcc,
4d3e0524fc59374f1fdad8e471c695469b45ecf69c1de85c6c1230e888dd4cbe,
36127f8aee7c61048984f0a208bf6d334db9dacbeeeef9ff2d17117e81232832
..1462eb3ef602f5911d77ab11f815eb4154ba95c934e414198ef000a61b4de31a,
755610da34cc224fbe60ce5e42add2ea6b272ef466aef18c13497363116d1c03,
d26274e014ebfc19a9c1a951193858b972eae3360ed35635e89f1f9dbe432be5,
26202144ba4c4cb7ecde831c9e9662bec519493b29a098dd5803a8b4d261fc12,
b9fa51c75d278d95f2ccace9609b28ec137b244c8b7d1523b16ed07c8e24b8e4,
e42423127a2ca12d4f199287c8fa07784eacf9fc9b86a6bd56ee364cc352c009,
5371d6f9c76b560b4e42b9154a395bed60924d8de31284e926d06af382f5ad1b,
```

Vector 5

```
da36359bf1bfd1694d3ed359e7340bd02a6a5e54827d94db1384df29f5bdd302,
decb0151cbeb49f76f10419ab6a96242bdc87baac8a474e5161123de4304ac29,
42616e646572736e6174636820766563746f72,
-,
-,
4315192d2ce9e52ceb449a6b4da7f7e6636e53592c7f5e236763e21e9bac24c7,
9508104b820469687488d83f729288d9f70fc0523318beff44a47da10d490b3c,
4ee61f3c000544aa48c565e143e05c6501a623bdbf02a0a408b97433660b4907
..715f75890cc0e45cdd7116e3da15b15c3c637782e8e05d05c0d5895e5fe583d1,
fb0123dd6317dbd379afccded247f75b3c1c2e32b86eaa9d6c9d0eb5bef07919,
a91807f0ee57d2344a8942808bf35c65b5bd4fde16752a98f3e3dc67be8c103d,
dbc69ea4dd299deec2f29845e98a17700f07842cf2f6c5e9d88f388fa3a2831c,
311f94e886825c80a30fd44535be37218501bd072afcbc1298f8fba6c3e3c96d,
15f1562046078a7c0d152ef1b56bf3078c763089bf08790f10ff1cf3b9a5030b,
```

```
f962cb2598032cf5b0a21b4c253514d75c91ace15acde9a3f716e40f70f06804,
```

Vector 6

```
da36359bf1bfd1694d3ed359e7340bd02a6a5e54827d94db1384df29f5bdd302,
decb0151cbeb49f76f10419ab6a96242bdc87baac8a474e5161123de4304ac29,
42616e646572736e6174636820766563746f72,
-,
1f42,
4315192d2ce9e52ceb449a6b4da7f7e6636e53592c7f5e236763e21e9bac24c7,
9508104b820469687488d83f729288d9f70fc0523318beff44a47da10d490b3c,
4ee61f3c000544aa48c565e143e05c6501a623bdbf02a0a408b97433660b4907
..715f75890cc0e45cdd7116e3da15b15c3c637782e8e05d05c0d5895e5fe583d1,
0752c5b639dffedf9a66ac111a765d3e9c4cfac9c8b26cc5af6d524967afdf0a,
d03caebf8577c1d2ed30a09708683195f11883411dc170e3ea9f09a2cbf86bab,
8b16f0abb2873d6d56199280aeee9e02ce0274a9ca06a3194d6a72c25516ace8,
311f94e886825c80a30fd44535be37218501bd072afcbc1298f8fba6c3e3c96d,
9671cdae8b4cdeea640c24993ccf7e571fcfb3344d81d3cc6f36d03496777c1c,
624e25cd6eccec59b09f0893ef9eab877b55c757b9e9c81260255145bffd9a0d,
```

Vector 7

```
35b877a25c394512292b82bdf8468e98eaf03c79c7fc9d53546dadc5fb75b500,
b0e1f208f9d6e5b310b92014ea7ef3011e649dab038804759f3766e01029d623,
42616e646572736e6174636820766563746f72,
-,
1f42,
4315192d2ce9e52ceb449a6b4da7f7e6636e53592c7f5e236763e21e9bac24c7,
6d1dd583bea262323c7dc9e94e57a472e09874e435719010eeafae503c433f16,
09106f062ac07846f3f841f64765527b333575143483855d633f99ccc2e8e306
..e6239ff79a1272cff931e8d0ac6c390328486329118ad40a18b85184da1837ff,
462ae9ad651e5caf11247b989fecb5f2b1729479c33b9133388d14fa35dbbd0c,
91f1ca92eeaa0b604faf3e4811c12b44991ea33cf582a529a4bc4429a3b6cc5a,
b69946f270c46ccc59557bd40288a0a27607281da1892328fdb9da2dcb6c73cf,
5a02419120b814a5c81d67096aac728ee9bda5ddf9451cf554d871462a04831a,
bd8c0c1e5e04577c8836e45fb64131d1275309fe28e1d4334b230e3aa639da1a,
d93ccbd393ed88c8165b0a01aabe28c56a53b43e527e7927eeadff006dd22114,
```

B.3. Ring VRF Test Vectors

KZG SRS parameters are derived from Zcash BLS12-381 powers of tau ceremony.

The evaluations for the ZK domain items, specifically the evaluations of the last three items in the evaluation domain $\mathbb{D}$, are set to 0 rather than being randomly generated.

Schema:

```
sk (x): Secret key,
pk (Y): Public key,
in (alpha): Input octet-string,
salt: encode_to_curve_salt (empty for this suite),
ad: Additional data octet-string,
h (I): VRF input point,
gamma (O): VRF output point,
out (beta): VRF output octet string,
blinding: Blinding factor,
proof_pk_com (Y^-): Pedersen proof public key commitment,
proof_r: Pedersen proof 'R' component,
proof_ok: Pedersen proof 'O_k' component,
proof_s: Pedersen proof 's' component,
proof_sb: Pedersen proof 's_b' component,
ring_pks: Ring public keys,
ring_pks_com: Ring public keys commitment,
ring_proof: Ring proof
```

Vector 1

```
3d6406500d4009fdf2604546093665911e753f2213570a29521fd88bc30ede18,
a1b1da71cc4682e159b7da23050d8b6261eb11a3247c89b07ef56ccd002fd38b,
-,
-,
-,
c5eaf38334836d4b10e05d2c1021959a917e08eaf4eb46a8c4c8d1bec04e2c00,
e7aa5154103450f0a0525a36a441f827296ee489ef30ed8787cff8df1bef223f,
fdeb377a4ffd7f95ebe48e5b43a88d069ce62188e49493500315ad55ee04d744
..2b93c4c91d5475370e9380496f4bc0b838c2483bce4e133c6f18b0adbb9e4722,
01371ac62e04d1faaadbebaa686aaf122143e2cda23aacbaa4796d206779a501,
3b21abd58807bb6d93797001adaacd7113ec320dcf32d1226494e18a57931fc4,
8123054bfdb6918e0aa25c3337e6509eea262282fd26853bf7cd6db234583f5e,
ac57ce6a53a887fc59b6aa73d8ff0e718b49bd9407a627ae0e9b9e7c5d0d175b,
0d379b65fb1e6b2adcbf80618c08e31fd526f06c2defa159158f5de146104c0f,
e2ca83136143e0cac3f7ee863edd3879ed753b995b1ff8d58305d3b1f323630b,
7b32d917d5aa771d493c47b0e096886827cd056c82dbdba19e60baa8b2c60313
..d3b1bdb321123449c6e89d310bc6b7f654315eb471c84778353ce08b951ad471
..561fdb0dcfb8bd443718b942f82fe717238cbcf8d12b8d22861c8a09a984a3c5
..a1b1da71cc4682e159b7da23050d8b6261eb11a3247c89b07ef56ccd002fd38b
..4fd11f89c2a1aaefe856bb1c5d4a1fad73f4de5e41804ca2c17ba26d6e10050c
..86d06ee2c70da6cf2da2a828d8a9d8ef755ad6e580e838359a10accb086ae437
..ad6fdeda0dde0a57c51d3226b87e3795e6474393772da46101fd597fbd456c1b
..3f9dc0c4f67f207974123830c2d66988fb3fb44becbbba5a64143f376edc51d9,
afd34e92148ec643fbb578f0e14a1ca9369d3e96b821fcc811c745c320fe2264
..172545ca9b6b1d8a196734bc864e171484f45ba5b95d9be39f03214b59520af3
..137ea80e302730a5df8e4155003414f6dcf0523d15c6ef5089806e1e8e5782be
```

```
..92e630ae2b14e758ab0960e372172203f4c9a41777dadd529971d7ab9d23ab29
..fe0e9c85ec450505dde7f5ac038274cf,
98bc465cdf55ee0799bc25a80724d02bb2471cd7d065d9bd53a3a7e3416051f6
..e3686f7c6464c364b9f2b0f15750426a9107bd20fe94a01157764aab5f300d7e
..2fcba2178cb80851890a656d89550d0bebf60cca8c23575011d2f37cdc06dcdd
..93818c0c1c3bff5a793d026c604294d0bbd940ec5f1c652bb37dc47564d71dd1
..aa05aba41d1f0cb7f4442a88d9b533ba8e4788f711abdf7275be66d45d222dde
..988dedd0cb5b0d36b21ee64e5ef94e26017b674e387baf0f2d8bd04ac6faab05
..7510b4797248e0cb57e03db0199cd77373ee56adb7555928c391de794a07a613
..f7daac3fc77ff7e7574eaeb0e1a09743c4dae2b420ba59cf40eb0445e41ffb24
..49021976970c858153505b20ac237bfca469d8b998fc928e9db39a94e2df1740
..ae0bad6f5d8656806ba24a2f9b89f7a4a9caef4e3ff01fec5982af8731433463
..62a0eb9bb2f6375496ff9388639c7ffeb0bcee33769616e4878fc2315a3ac351
..8a9da3c4f072e0a0b583436a58524f036c3a1eeca023598682f1132485d3a570
..88b63acd86c6c72288568db71ff15b7677bfe7218acdebb144a2bf261eb4f659
..80f830e77f37c4f8d11eac9321f302a089698f3c0079c41979d278e8432405fc
..14d80aad028f79b0c4c626e4d4ac4e643692a9adfdc9ba2685a6c47eef0af5c8
..f5d776083895e3e01f1f944cd7547542b7e64b870b1423857f6362533f7cd2a0
..1d231ffed60fe26169c28b28ace1a307fdc8d4b29f0b44659402d3d455d719d8
..96f83b7ee927f0652ca883e4cfa85a2f4f7bc60dda1b068092923076893db5bd
..477fa2d26173314d7512760521d6ec9f,
```

Vector 2

```
8b9063872331dda4c3c282f7d813fb3c13e7339b7dc9635fdc764e32cc57cb15,
5ebfe047f421e1a3e1d9bbb163839812657bbb3e4ffe9856a725b2b405844cf3,
0a,
-,
-,
8c1d1425374f01d86b23bfeab770c60b58d2eeb9afc5900c8b8a918d09a6086b,
60f32f5ad3e9694b82ccc0a735edb2f940f757ab333cc5f7b0a41158b80f574f,
44f3728bc5ad550aeeb89f8db340b2fceffc946be3e2d8c5d99b47c1fce344b3
..c7fcee223a9b29a64fe4a86a9994784bc165bb0fba03ca0a493f75bee89a0946,
99ff52abf49d67c4303ac4a8a00984d04c06388f5f836ebd37031f0e76245815,
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Vector 6

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